

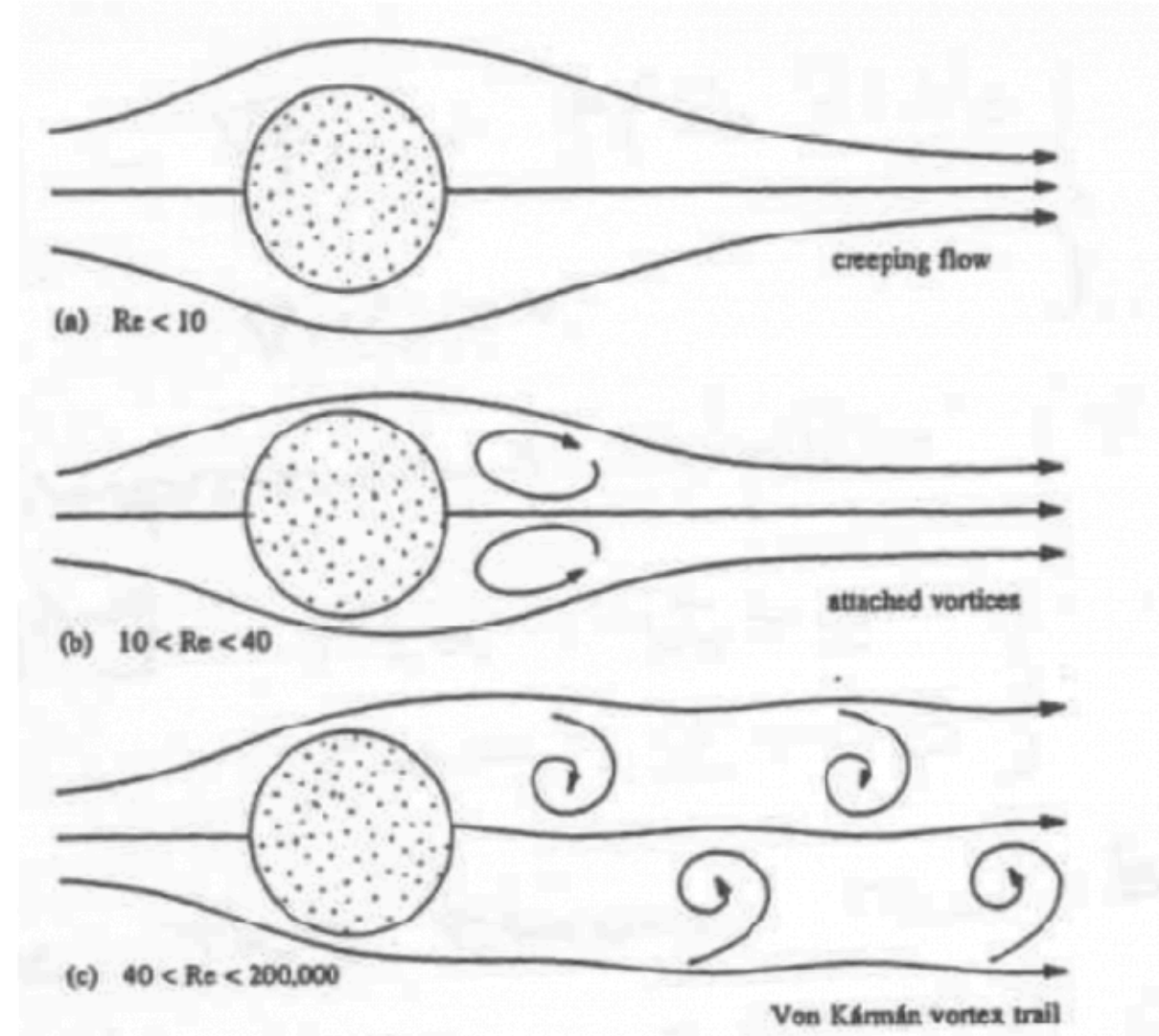
Day 06 - Making Classical Models



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Plane Polar Coordinates Warm-Up

We introduced plane polar coordinates (r, ϕ) . For any position vector, $\vec{R}$, we can write:

$$\vec{R} = |\vec{R}| \hat{r} = r \hat{r}$$

where r is the magnitude of $\vec{R}$, and $\hat{r}$ is the radial unit vector.

Find $\dot{\vec{R}} = \frac{d\vec{R}}{dt}$. Get as far as you can. Our answer will be in terms of $\hat{r}$ and $\hat{\phi}$.

Remember the chain rule and Cartesian unit vectors are fixed in space/time

$$\begin{aligned} \hat{r} &= \cos(\phi) \hat{x} + \sin(\phi) \hat{y} & \hat{\phi} &= -\sin(\phi) \hat{x} + \cos(\phi) \hat{y} \\ \frac{d}{d\phi} \cos \phi &= -\sin \phi & \frac{d}{d\phi} \sin \phi &= \cos \phi \end{aligned}$$

Announcements

- Homework 2 is due Friday (late Sunday)
- Homework 3 is now posted
- Office hours, for now (Mihir-MN; Danny-DC):
 - Friday 10-12pm (DC, 1248 BPS)
- Zoom Link for class: <https://msu.zoom.us/j/99550311023>
 - password: phy321msu

Seminars this week

WEDNESDAY, January 28, 2026

Astronomy Seminar, 1:30pm, BPS 1400 & Zoom

Speaker: Rebecca Kyer & Emily Elizondo, MSU

Title: TBA

Zoom Link: [https://msu.zoom.us/j/93334479606?](https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1)

[pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1](https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1)

Passcode: 825824

Seminars this week

WEDNESDAY, January 28, 2026

Physics Education Research Seminar, 3:00pm, BPS 1400 & Zoom

Speaker: Jennifer Doherty, MSU

Title: Principle-based reasoning: A strategy for developing expertise in Physiology

Zoom Link: <https://msu.zoom.us/j/96470703707>

Passcode: PERSeminar

Seminars this week

WEDNESDAY, January 28, 2026

FRIB Nuclear Science Seminar, 3:30pm, Zoom Only

Speaker: Gwen Grinyer, University of Regina

Title: Precision Spectroscopy of Rare Isotopes

Zoom Link: [https://msu.zoom.us/j/91051885898?](https://msu.zoom.us/j/91051885898?pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1)

[pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1](https://msu.zoom.us/j/91051885898?pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1)

Passcode: 949110

Seminars this week

FRIDAY, January 30, 2026

QuIC, Seminar, 12:40 p.m., BPS 1300 & Zoom

Speaker: Ben DalFavero, MSU

Title: Fault tolerant quantum computing II

*For the full schedule, please see: <https://sites.google.com/msu.edu/quic-seminar/> or for more information, please reach out to Ryan LaRosa directly

Seminars this week

FRIDAY, January 30, 2026

IReNA Online Seminar, 2:00 pm, Zoom

Light refreshments at 1:50pm in 2025 Nuclear Conference Room - FRIB

Hosted by: Aldana Grichener (University of Arizona & Observatory)

Speaker: Mengke Li, University of California, Berkeley

Title: Implications of a Weakening $N = 126$ Shell Closure Away from Stability
for r-Process Astrophysical Conditions

Zoom Link: <https://msu.zoom.us/j/827950260>

Password: CENAM

Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Modeling Video



Source: <https://www.youtube.com/watch?v=dkTncoPqo5Y>

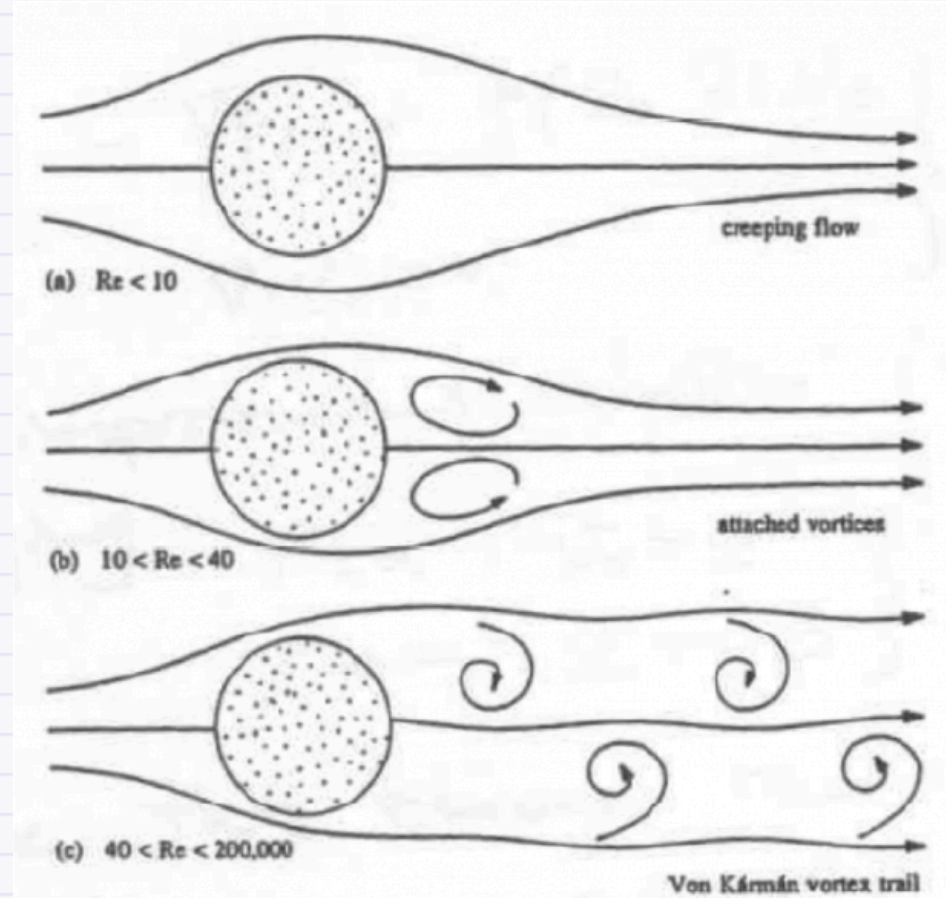
What is your experience with modeling?

Take 2-3 min to think about your prior physics classes

- What models have you used? What makes that a model?
- What made a that model good or not so good?
- What kinds of things could you do to make a better model?

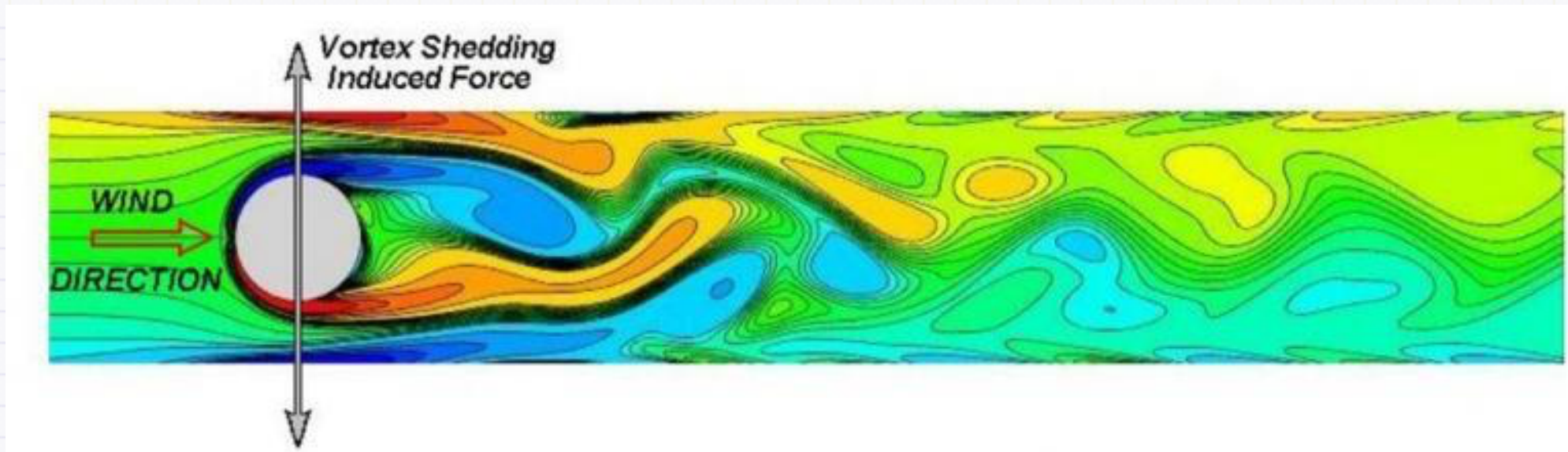
Vortex Shedding

- At higher Reynolds numbers, flow around objects becomes unstable.
- This instability can lead to the formation of vortices.
- This "shedding" of vortices can lead to vibrations and noise.



Model of vortex shedding behind a cylinder

- Controlling vortex shedding is important in many engineering applications.



Renewables: Wind Turbines

Thorntonbank Wind Farm

North Sea off the coast of Belgium

Notice the cylindrical shape of the support structure.



Clicker Question 6-1

The SHO is a useful model: $m\ddot{x} = -kx$.

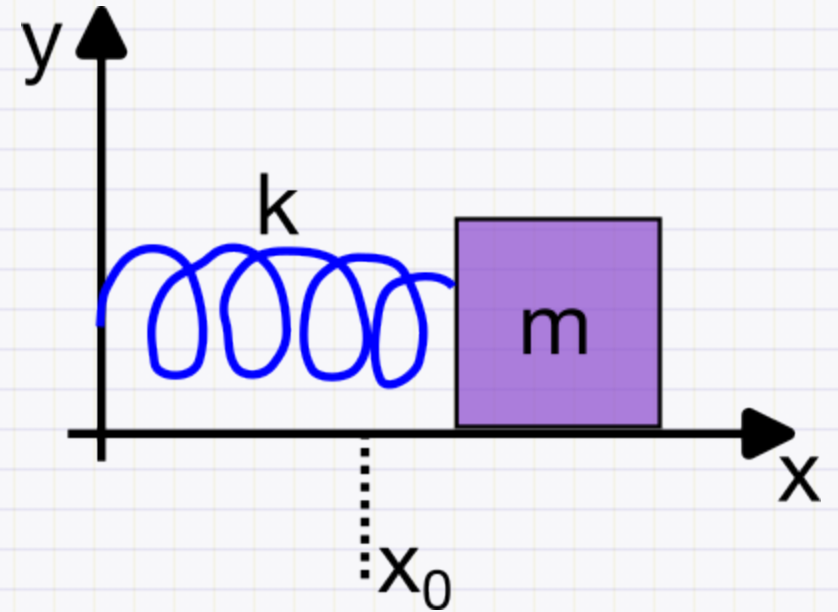
Assume the **restoring force is anti-symmetric** about the equilibrium position, what is the next term model?

1. $\sim x^2$

2. $\sim x^3$

3. $\sim x^4$

4. $\sim x^5$



Clicker Question 6-2

Assuming a **linear model** for Air Resistance $\sim bv$, we obtained this EOM for a falling ball:

$$\ddot{y} = -g + \frac{b}{m}\dot{y}$$

What happens when $\ddot{y} = 0$?

1. The ball stops moving ($v = 0$).
2. The ball reaches a velocity of mg/b .
3. The ball reaches a terminal velocity.
4. I'm not sure.

Clicker Question 6-3

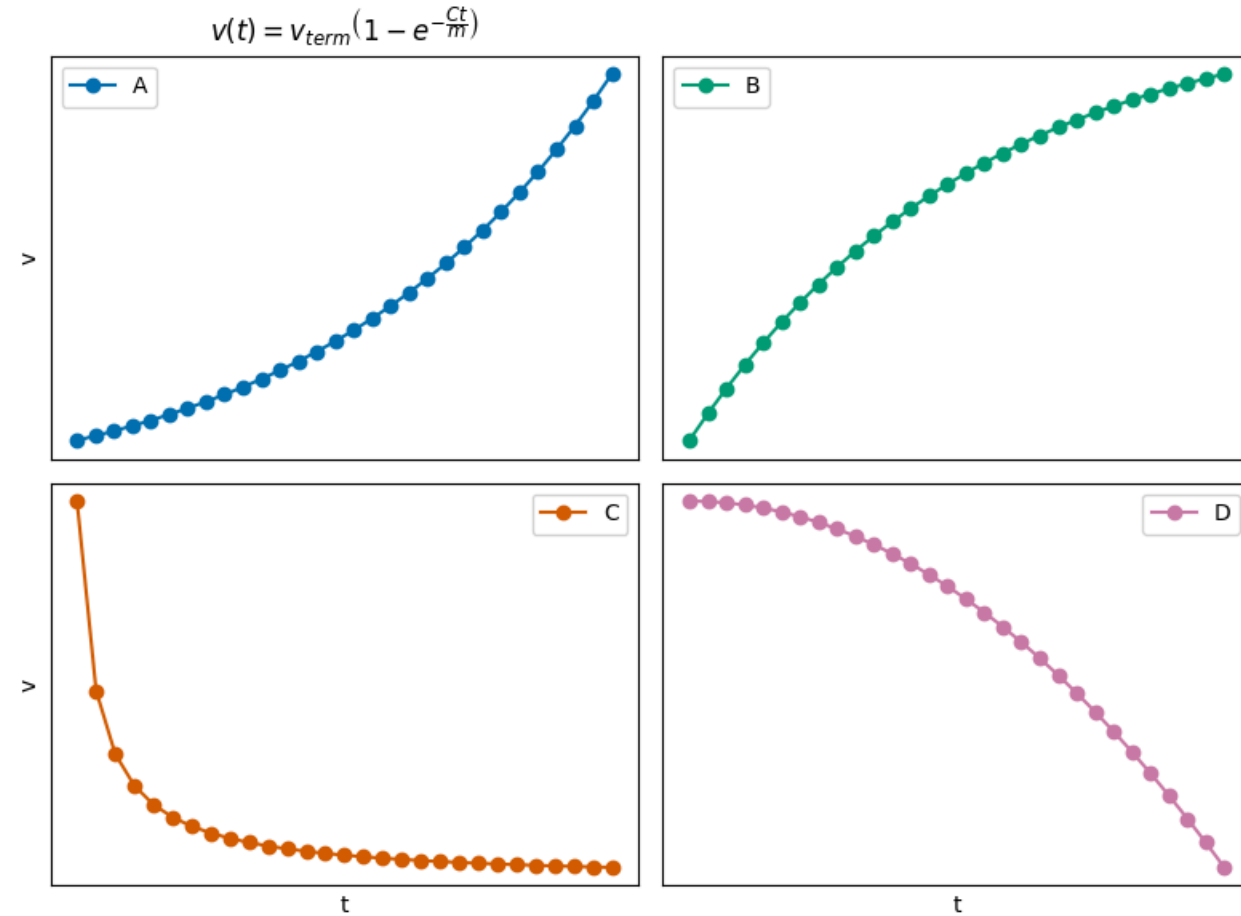
For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units $[gt/v_{term}]$?

1. Yes, v and $(mg/c)^{1/2}$ **have the same units**; the units for $[gt/v_{term}]$ are m/s.
2. No, v and $(mg/c)^{1/2}$ **have different units**; the units for $[gt/v_{term}]$ are m/s.
3. Yes, v and $(mg/c)^{1/2}$ **have the same units**; the units for $[gt/v_{term}]$ are unit-less.

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.